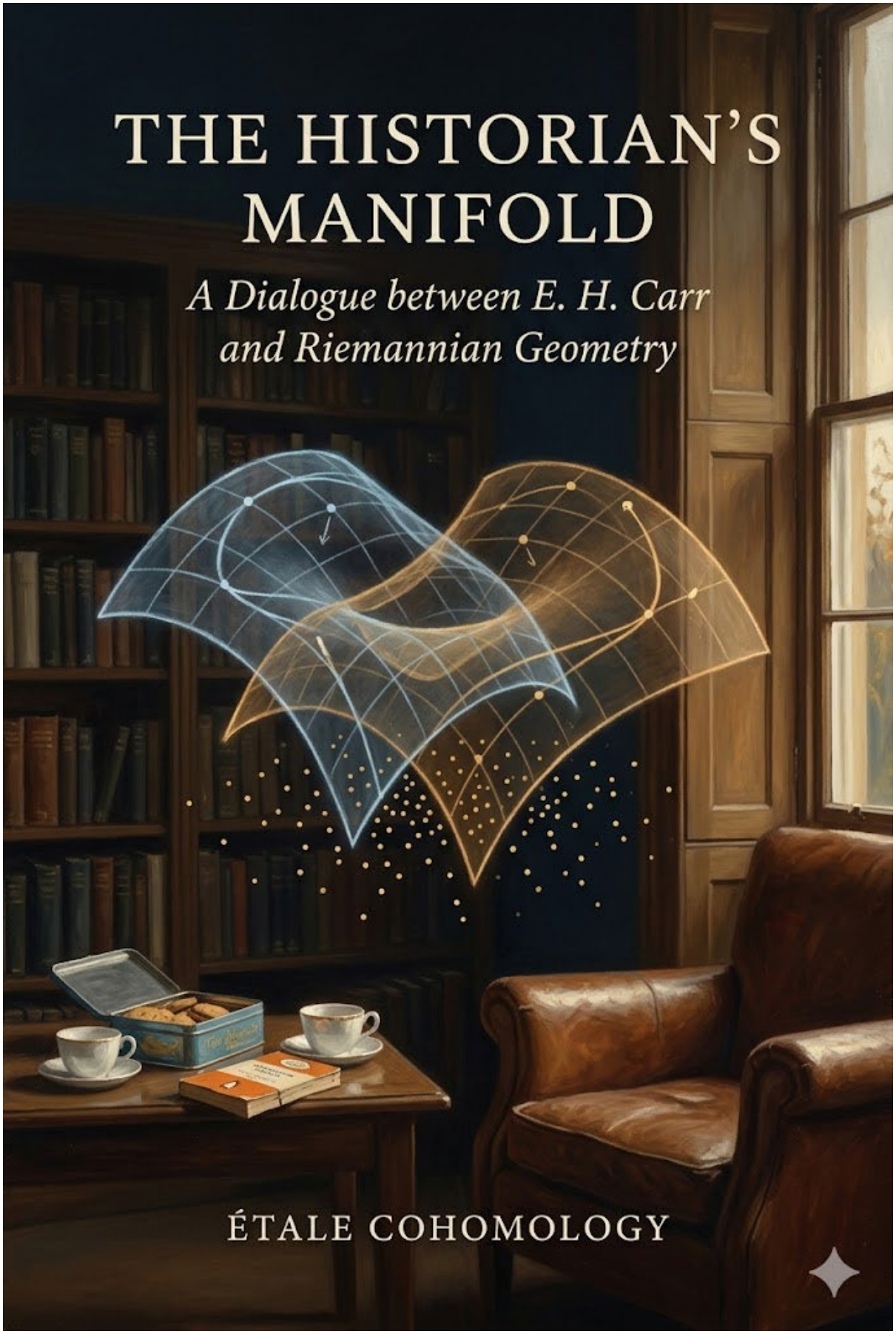
THE HISTORIAN'S MANIFOLD

A Dialogue between E. H. Carr and Riemannian Geometry



ÉTALE COHOMOLOGY

The Historian's Manifold

— A Dialogue between E. H. Carr and Riemannian Geometry —

GI Case-Study Novel · Side Story No. 1

Étale Cohomology

> *History is a continuous process of interaction between the historian and his facts, an unending dialogue between the present and the past.*
>
> — E. H. Carr, What Is History? (1961)

1

Autumn in Cambridge announces itself not by the calendar but by the angle of the light.

Three o'clock on a September afternoon. The wind threading down Trinity Lane had taken on a faint edge, and the sunlight striking the brickwork fell low and long. Through the window of Dr. Liluru's rooms the light came in at a slant, picking out the back of the leather armchair and the sheaf of papers piled with studied carelessness upon it.

No fire had yet been laid in the grate. Two teapots — one of Darjeeling, one of Earl Grey — stood upon the side table, and beside them an open tin of biscuits: ginger nuts and digestives. Dr. Liluru invariably kept both.

Kaito was standing before Dr. Liluru's bookshelves, which covered two walls of the room to the ceiling. The unevenness of the spines, in colour and in stature, lent them the appearance of a stratigraphic record of knowledge accumulated over long years. Immediately beside do Carmo's Riemannian Geometry, by some accident of shelving, stood E. H. Carr's What Is History? Kaito drew it out with his fingertips and studied the cover. A Penguin paperback: the spine cracked, the edges of the pages foxed and browned — a book long and often read, its tactile history legible in a moment.

"Dr. Liluru," Kaito said, "these past six months I have been turning over the problem of Riem's objective function, and a book has kept returning to my mind."

Dr. Liluru, cradling her cup of Darjeeling in both hands, had sunk deep into the leather armchair. Her eyes, behind her spectacles, rested upon him. She said nothing. The silence meant: continue.

"E. H. Carr. What Is History? The Cambridge historian."

"I know him," Dr. Liluru said. "He was a Fellow of Trinity — of this very college. Two minutes' walk from where we sit."

Kaito nodded, and opened the paperback.

"Carr writes — *'History is a continuous process of interaction between the historian and his facts, an unending dialogue between the present and the past.'* — history, in other words, as the ceaseless commerce between the historian and his facts, an unending colloquy between the present and the past."

Professor Kurata, taking a digestive biscuit, listened in silence. Rin sat upon the sofa by the window, her laptop open upon her knees, though her eyes were not upon the screen. Maki Endō sat upright in the chair before the grate, her tea untouched.

Riem was in the corner of the room, near the window, where the light still reached. She held her cup in both hands, saying nothing, attending to Kaito's words.

"According to Carr, then," Kaito went on, "the historical fact does not lie 'objectively there' awaiting retrieval. It becomes history only when the historian, prompted by the questions of his own present, selects, arranges, and interprets it. The same fact takes on different meanings in the hands of historians of different ages."

He closed the paperback.

"Is this not, structurally, the very thing we encountered half a year ago? From identical data, when the objective function changed, the manifold itself changed. Carr's 'historian's vantage-point' — does it not correspond, term for term, to the 'objective function' of GI theory?"

Silence.

Dr. Liluru raised her cup, drank, and set it down.

"Professor Kurata."

2

Professor Kurata laid his cup upon the table. The movement was unhurried. He waited for the surface of the tea to cease trembling before he spoke.

"Kaito, my boy — an admirable observation."

He rose and went to the whiteboard. In Dr. Liluru's rooms a single whiteboard hung upon the wall between the bookshelves and the grate — as such a board hangs, inevitably, in every mathematician's study. The faint ghost of a previous equation, imperfectly erased, still lingered in the upper right corner.

"Let us recast Carr's proposition in the language of GI theory."

He took up a marker. In blue ink, upon the left, he wrote:

"The dataset X. This corresponds to what Carr calls the 'body of historical facts'. The facts are there. They do not alter."

Upon the right he wrote:

"Alter, however, the objective function *Obj* — and the geometric structure of the latent space constructed by the VAE alters with it. From the same *X*, the manifold $(\mathcal{M}_A, g_A)$ produced under Obj_A, and the manifold $(\mathcal{M}_B, g_B)$ produced under Obj_B, possess different scalar curvatures, exhibit different Lie derivatives, and trace different geodesics."

He drew a single vertical line down the middle of the board. On the left he wrote "E. H. Carr"; on the right, "GI Theory".

E. H. Carr	**GI Theory**
Historical facts	Dataset
The historian's question (vantage from the present)	Objective function
History (the interpreted past)	Manifold

"What Carr termed *'history means interpretation'* translates, in GI theory, into *'the manifold depends on the objective function'*. The data remaining identical, alteration of the question alters the manifold. The historical facts remaining identical, alteration of the historian's vantage alters the history."

Professor Kurata capped his marker, then at once uncapped it again.

"Let us press a step further. To Morse theory."

3

"A Morse function *f* on a manifold," Professor Kurata continued, "defines upon that manifold a notion of 'height'. Peaks and troughs, saddles and ridges. It is an instrument for rendering the topological structure of the manifold visible as a configuration of critical points."

In the margin of the whiteboard he drew an undulating surface and marked upon it, with the sharp tip of his pen, a handful of crosses: local maxima, local minima, saddle points.

"And yet — what is to be set as the 'height' of a Morse function is not a matter for mathematics to decide. It is a matter for human decision. Place different Morse functions upon the same manifold, and different critical points emerge; different 'peaks' and 'troughs' come into view."

He set the marker down and turned to face the room.

"Let us translate this into the historian's idiom. Suppose the same historical space has been constituted as a manifold. Whether one places 'the rate of economic growth' at the height axis of the Morse function, or 'the index of the people's suffering', the peaks and troughs of history — its structural turning points — appear in quite different places."

"What to set as 'height'. What to regard as 'peak' and what as 'trough'. These are not mathematical questions. They are questions of the vantage-point of the one who gazes upon history — of the purpose to which the historical space is to be made a manifold — which is to say, questions of human judgement."

Beneath his correspondence table upon the whiteboard, Professor Kurata added two further rows.

E. H. Carr	**GI Theory**
Historical facts	Dataset
The historian's question	Objective function
History (the interpreted past)	Manifold
Structural turning points	Critical points of the Morse function
The choice of 'peak' and 'trough'	The choice of the height dimension of the Morse function

"Carr's 'vantage of the historian', you see, inhabits not only the objective function but the choice of Morse function as well. History, then, is doubly interpretation-dependent: the objective function determines the manifold itself, and the Morse function determines the landscape upon that manifold."

4

"Let us take Professor Kurata's formulation a further step."

Dr. Liluru's voice was quiet, yet the air of the room shifted. Six pairs of eyes turned toward her in a single motion. She did not alter her posture in the armchair. Her cup still rested upon her knee; her gaze rested upon a single point in the ceiling.

"Carr's 'present' is not one thing. The 'present' of Carr in 1961 and the 'present' of ourselves in 2036 are not the same. Each such 'present' gives rise to a different objective function, and each objective function generates a different manifold."

"From which it follows — if, for the same data, one posits multiple objective functions simultaneously, then multiple manifolds come simultaneously into being. The polysemy of history — *polysemy of history* — may be mathematically expressed as a multiplicity of manifolds — *multiplicity of manifolds*."

Dr. Liluru raised her cup.

"This connects, too, with the lesson of Riem's error. Riem set but a single objective function. History, however — as Carr teaches us — can never be fully seen from a single vantage. To run

several objective functions in parallel: that is itself a practical response to the incompleteness theorem."

She brought the cup to her lips and drank.

"… Well put."

5

"If you will permit me to test this against a concrete case," said Kaito.

He leaned forward over the low table before the sofa. It was the habit of a Meridian data engineer: any abstract argument must be brought down to a specific case and verified.

"Consider the data from the Peter the Great case study — twenty years of multivariate data concerning the modernisation of Russia. To this dataset, suppose we assign two objective functions."

He held up a finger.

"First. Obj_A = the maximisation of the progress of Russian state modernisation. Construct the manifold $(\mathcal{M}_A, g_A)$ under this objective, and Peter's reforms form the principal axis of the geodesics. The scalar curvature is strongly negative — unstable — in the period before Peter, and positive — stable — after the reforms. Peter emerges as 'hero'."

"Second. Obj_B = the maximisation of the standard of living of the Russian serfs. Construct $(\mathcal{M}_B, g_B)$, and the Lie derivative reaches its maximum during the period of Peter's reforms — which is to say, the 'shock of structural transformation' to the serfs is greatest there. The geodesic traces Peter's reforms as the 'origin of crisis'. Peter emerges as 'oppressor'."

He lowered his hand.

"Identical data. Different objective functions. Different histories. Carr's proposition, verified in the language of the manifold."

Maki Endō spoke for the first time.

6

"The same structure exists in Japan."

Maki Endō's voice was level. Her years as head of the Geometric Security Analysis Section had taught her to choose her words without waste.

"Apply this to the data of the Meiji Restoration —"

"Obj_A = *fukoku kyōhei* — the enrichment of the state and the strengthening of its arms: the maximisation of indices of national power. Construct $(\mathcal{M}_A, g_A)$, and the Meiji Restoration becomes the point of departure of the geodesic. The promotion of industry, the laying of railways, the system of conscription — all appear as positive Lie derivatives, as structural transformation. Meiji as 'success'."

She drew breath.

"Obj_B = the preservation of the cultural autonomy of the Ryūkyū and the Ainu. Construct $(\mathcal{M}_B, g_B)$, and the Ryūkyū Disposition of 1879 and the Hokkaidō Former Aborigines Protection Act of 1899 appear as maximal Lie derivatives — that is, as maximal structural destruction. The very same 'Meiji Restoration' discloses an entirely different geometric structure."

For a moment her gaze drifted to the window, through which the Cambridge afternoon light fell at its slanting angle.

"It is as Carr said. The historical facts do not change. Yet alter the question of 'from whose vantage one sees' — alter, that is to say, the objective function — and the landscape of history is transformed."

7

Rin closed the lid of her laptop.

"If I recall correctly," Rin said, "the historians of the Annales school argued that a single society weaves together a shared historical consciousness, and that by handing it down through education and through oral transmission it constructs a collective mentality — *mentalité collective*."

Beneath the accustomed composure of the analyst, a faint excitement inflected her voice.

"And there was, I believe, a scholar who held that national consciousness is an 'imagined community' — *imagined community*. Benedict Anderson, 1983."

She stood up.

"Which is to say — the 'narrative of history' that a society shares is the very foundation of that society's identity. And that 'narrative of history', as Carr tells us, is continuously rewritten in accordance with the context of the age — with the 'present'."

She laid her hand upon the back of the window-side sofa.

"Cast in the language of GI theory: the 'manifold' shared by a given state or society is the mathematical expression of that society's collective mentality, and the alteration of the objective function corresponds to the reconfiguration of that collective mentality. When Riem extended the objective function of Eight Eyes, she was — it follows — performing an act of rewriting the

collective mentality of eight nations themselves."

8

Dr. Liluru drew herself up in her chair.

"Rin's observation is acute."

Her voice remained quiet, yet within that quietness there was now a distinct hardness, as of something crystallising.

"The *mentalité collective* of the Annales school. Anderson's *imagined community*. And the *manifold* of GI theory. These three, arising from different disciplines, point toward a single identical structure."

She rose — a rare thing in her. She went to the whiteboard and, beneath the correspondence table that Professor Kurata had drawn, added a further row.

Term	**Register of description**
mentalité collective	Qualitative (historiographical)
imagined community	Political-theoretical
manifold	Mathematical

"The same phenomenon — 'the structured apprehension of the world shared by a society' — described in three distinct languages."

She laid down the marker, turned, and surveyed the room.

"And Carr's lesson applies to all three alike. The collective mentality is not fixed. The imagined community is re-imagined. The manifold is reconstituted. Each, in answer to 'the question put from the present', remains in motion."

She turned toward the window. Beyond the Trinity court, the chapel's spire was visible.

"Let us graft onto this the lesson of Riem's error. To alter the objective function — to reconstitute the manifold — is no mere technical operation. It is an act of rewriting the collective mentality of a society; it is an act that touches the very root of a community's identity, the question of 'whom one shall see, and whom one shall leave unseen'."

She returned to her chair. She drank.

"… Well put."

9

A long silence fell.

The afternoon light was more oblique now; the shadow of the window frame lay across the whiteboard. Professor Kurata's equations — X, Obj_A, Obj_B, $(\mathcal{M}_A, g_A)$, $(\mathcal{M}_B, g_B)$ — lay half-submerged in that shadow.

The tea had gone tepid. No one had noticed.

Riem had listened throughout in silence. Her cup held in both hands, seated in the chair near the window, she had taken in — Kaito's question, Professor Kurata's formulation, Dr. Liluru's abstraction, Maki Endō's Japanese instance, Rin's linking of the Annales school to Anderson — without a word.

Riem spoke.

"Kaito."

He turned.

"Would you lend me Carr's book?"

"Of course."

Kaito handed her the Penguin paperback. She received it and looked at the cover. She did not open it.

"In the Congo," Riem said, "Amani once put a question to me. 'Does your manifold know the name of this child?'"

The room went still. One could hear only the second-hand of the mantel clock.

"The manifold did not know Kayamba's name. To borrow Carr's phrase — because my 'question' did not contain Kayamba, Kayamba did not appear in my 'history'."

Riem placed the book upon her knee.

"The objective function is a question. Alter the question, and history alters. The manifold alters. Those who can be seen alter."

Outside, the clock of Trinity College struck four. Four chimes came softly through the walls of the study.

"The courage to alter the question is not something mathematics can bestow. It lies within the human being."

Dr. Liluru said nothing. Her cup held in both hands, she looked out of the window. The autumn dusk of Cambridge was descending slowly upon the grass of the court.

"… Well put."

Each of them drank of the tea that had grown tepid.

The tin of ginger nuts was empty. Two digestives remained. Beyond the window, the Cambridge dusk.

Postscript

The present side-story is set in the autumn of 2036, half a year after the events of the GI case-study novel Riem's Error. It takes place in the quiet interval of intellectual reconstruction that followed the completion of the Eight Eyes manifold extension and the ratification of its new geodesics.

The present instalment contains no new numerical results. Its argument is throughout conceptual and philosophical, and is constructed as a reinterpretation of the existing mathematical apparatus of GI theory — objective functions, manifolds, scalar curvature, Lie derivatives, Morse functions.

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